

Embodied AI: Bridging Simulation and Reality in Robotics

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Abstract—This paper presents a comprehensive review of simulation-to-reality (Sim2Real) transfer techniques in the context of embodied artificial intelligence (embodied AI). Embodied AI refers to artificial intelligence systems that interact with the environment through a physical or virtual body (e.g., robots, virtual agents), enabling perception, decision-making, and action in real or simulated spaces. We focus on their applications in robotic learning and control. By critically comparing the capabilities of mainstream simulation platforms such as Habitat and Isaac Gym, the study identifies core trade-offs between physical realism and computational efficiency. It further analyzes transfer methodologies, including domain randomization, domain adaptation, and hybrid techniques, highlighting their effectiveness and limitations in dynamic and unstructured environments. The review also investigates the emerging role of large multimodal models, such as PaLM-E and RT-2, in bridging semantic understanding with robotic action. While these models offer significant improvements in generalization and task planning, challenges remain in achieving real-time performance and physical grounding. The findings suggest that simulation fidelity alone does not guarantee successful transfer, hybrid transfer methods outperform single-strategy approaches in complex settings, and large language models hold promise for enhancing robot intelligence but must be optimized for embedded deployment. This study offers practical insights for developing scalable and cost-effective Sim2Real pipelines in industrial applications such as autonomous navigation and robotic manipulation. It further outlines future directions in edge computing, lightweight modeling, and ethics-aware simulation, promoting the integration of physics-aware AI in real-world robotic systems.

Keywords—*Sim2Real, Embodied AI, Simulation Platforms, LLMs, Sim2Real Transfer Methods*

I. INTRODUCTION

Recent advancements in Embodied Artificial Intelligence (Embodied AI) have opened new frontiers in robotics. High-fidelity simulation platforms such as Habitat [1] and Isaac Gym [2] have revolutionized robotic learning by providing scalable, cost-effective training environments. However, the inherent simulation-to-reality (Sim2Real) gap remains a critical challenge, limiting the direct application of simulation-trained models in dynamic, unstructured real-world scenarios. Discrepancies in physics modeling, sensor noise, and environmental variations collectively contribute to this gap, necessitating the development of robust transfer learning methodologies.

Despite significant progress, current research still faces key limitations. Many Sim2Real methods require extensive

real-world data, contradicting the goal of cost-efficient training. Moreover, most simulation platforms struggle to balance physical realism with computational scalability, which hinders the training of robust models in diverse environments. The emergence of large language models (LLMs) and multimodal foundation models (e.g., PaLM-E [3], RT-2 [4]) offers a promising paradigm shift by integrating linguistic reasoning with perceptual inputs, significantly enhancing task generalization. However, their high computational demands and lack of physical embodiment present practical barriers for real-time, real-world applications.

The importance of this topic is underscored by its wide-ranging implications across industries such as autonomous driving, warehouse automation, and assistive robotics, where reliable Sim2Real transfer is essential to reduce dependency on expensive and labor-intensive data collection. As robotic systems are increasingly expected to function autonomously in complex and uncertain environments, the ability to transfer learning from simulation to reality becomes a foundational requirement for scalable and intelligent deployment.

This review aims to systematically analyze the current state of Sim2Real transfer in embodied AI with the following goals: First, it compares the technical characteristics and limitations of modern simulation platforms; it also evaluates the effectiveness of various transfer learning methods in bridging the Sim2Real gap; and it examines how embodied large models are revolutionizing robotic learning paradigms. By addressing these objectives, this review provides a comprehensive framework for selecting appropriate tools and methodologies, offering practical insights to both researchers and industry practitioners.

Ultimately, this study contributes to advancing the practical application of embodied AI by identifying optimal strategies for simulation-based training and transfer. It also highlights future directions, such as developing lightweight models for edge deployment and designing physics-aware simulation environments, to help enable intelligent robots that can operate effectively and reliably in diverse real-world scenarios.

II. CURRENT RESEARCH PROGRESS

In recent years, significant progress has been made in simulation platforms and transfer learning methods for robotic learning. Platforms like Habitat [1] and Isaac Gym [2] offer efficient simulation environments, but there are still trade-offs between physical accuracy and real-time performance. Domain Randomization (DR) [5] and hybrid transfer methods

[6] show superiority in both static and dynamic environments, but challenges remain regarding reliance on real-world data and lack of standardized evaluations.

Embodied large models like PaLM-E [3] and RT-2 [4] enhance cross-task generalization, but limitations still exist in real-time control and learning new skills. The trade-off between large and lightweight models, as well as the reduced reliance on real-world data through multimodal pretraining, remains a central issue in current research.

A. Simulation Platforms and Their Evolution

In the context of embodied AI, simulation platforms for robotic learning refer to software environments that mimic physical and perceptual aspects of the real world, enabling virtual training of robotic agents. These platforms allow robots to learn, plan, and interact with simulated environments before being deployed in real-world scenarios, thereby reducing the cost, risk, and time associated with physical trials.

Recent years have witnessed significant advancements in such simulation platforms. Habitat [1] introduced photorealistic 3D indoor environments specifically designed for embodied AI research, enabling vision-and-language navigation tasks under visually realistic conditions. Its successor, Habitat 2.0 [7], further extended the platform by incorporating dynamic object interaction, allowing for more complex tasks such as rearrangement and manipulation. Meanwhile, Isaac Gym [2] transformed physics simulation by leveraging GPU-based parallelism to support over 10,000 environments simultaneously, making it ideal for large-scale reinforcement learning.

Comparative studies [8] have revealed fundamental trade-offs among simulators. For instance, MuJoCo [9] offers high-fidelity simulation of contact dynamics, making it well-suited for fine-grained manipulation tasks. In contrast, PyBullet prioritizes computational efficiency, enabling faster iterations and broader scalability for training multiple agents in parallel.

However, these platforms still face inherent limitations. As demonstrated in [8], no single simulator can currently achieve both high physical accuracy and real-time performance in complex, interactive settings. Additionally, datasets such as HM3D [10], which aim to increase environment diversity for navigation tasks, still lack essential real-world features like material properties and sensor noise, limiting their generalizability in practical deployment scenarios.

B. Transfer Learning Methodologies

Sim-to-Real (Sim2Real) transfer refers to the process of applying models trained in simulated environments to real-world robotic systems. Since simulations offer a safe, controllable, and cost-effective environment for training, transferring these learned policies or perception models to real hardware is essential for practical deployment. However, due to discrepancies between simulation and reality, this transfer remains a major challenge in embodied AI.

Domain randomization (DR) [5] has emerged as a dominant strategy for sim-to-real transfer, following the seminal work by Tobin et al. [5], which demonstrated successful vision-based grasping policy transfer by

randomizing textures, lighting, and camera parameters. Peng et al. [11] extended DR to dynamics, improving policy robustness to variations in robot mechanics. However, as Tzeng et al. [12] highlighted, pure DR often struggles under substantial domain shifts, prompting the development of hybrid approaches that combine DR with domain adaptation or representation learning techniques.

Adversarial domain adaptation (ADA) methods [12] have shown promise by explicitly reducing the feature distribution discrepancy between simulated and real domains through adversarial training. Building on this, the Sim-to-Real via Sim-to-Sim framework [13] introduced intermediate simulation domains that progressively bridge the reality gap.

Current limitations in sim-to-real transfer research include (1) most approaches rely on access to some real-world data [12], which contradicts the ideal of zero-shot transfer; (2) dynamic environments, such as those involving human-robot interaction, remain challenging due to the unpredictable behavior of external agents [11]; and (3) existing benchmarks [10] lack standardized evaluation metrics, making cross-method comparisons difficult and potentially unreliable.

The integration of large language models has fundamentally altered Sim2Real paradigms. PaLM-E [3] achieved remarkable cross-task generalization by grounding language instructions in visual and robotic action spaces. For instance, it could execute "pick up the green block" commands in novel environments after only simulation training [3]. RT-2 [4] further demonstrated knowledge transfer from web-scale data, enabling robots to interpret abstract commands.

However, critical challenges persist: As Yu et al. [14] demonstrated, when tasks require interaction with specific parts of a scene that are hard to describe verbally, relying solely on natural language becomes restrictive. As D. Driess et al. [3] demonstrated, while the method is general and flexible, it has limitations in motion augmentation, temporal consistency, and real-time efficiency, which could be addressed through simulation data and more efficient models. Although RT-2 [4] shows strong generalization, it has key limitations: it doesn't learn new physical skills beyond its training data, high computational cost may hinder real-time control, and there's a lack of accessible VLM models.

C. Contradictions and Challenges in Embodied AI

While conventional approaches prioritize high-fidelity physics simulation (e.g., using MuJoCo [9] for precise contact modeling), empirical evidence shows that low-dimensional semantic representations (e.g., visual features extracted by ResNet) often enable better cross-scene transfer than detailed physical modeling, particularly in tasks like visual navigation. Although larger models such as PaLM-E [3] demonstrate superior knowledge capacity and semantic reasoning, smaller, lightweight models like DistilBERT or TinyLLaMA exhibit better operational efficiency and adaptability in real-time control scenarios, especially on resource-constrained robots. Multimodal pretraining (e.g., vision-language pairing in RT-2 [4]) significantly reduces reliance on real robotic data, yet achieving fully end-to-end training without any real-world

interaction remains a major challenge, particularly in physically interactive tasks.

III. AI EMBODIMENT PLATFORMS AND APPROACHES

A. Abbreviations and Acronyms

Recent advancements in simulation platforms have revealed significant performance trade-offs, particularly in the context of robotic learning. High-fidelity simulators, such as MuJoCo [9], are widely recognized for their superior physical realism, which makes them ideal for tasks that require precise contact dynamics and physical accuracy. However, this realism comes at a cost, as MuJoCo incurs substantial computational overhead, limiting its scalability for large-scale training environments.

In contrast, platforms like PyBullet and Isaac Gym [2] leverage GPU acceleration to enable faster, parallelized simulations. This allows them to handle large-scale environments with improved computational efficiency. Although this parallelization enhances scalability, it does so at the expense of slightly reduced physical precision compared to high-fidelity simulators.

Habitat [1], a platform designed to focus on photorealistic rendering, excels in tasks related to navigation and perception. However, it has limitations in supporting dynamic object interactions, which makes it less suited for tasks requiring physical manipulation. On the other hand, Isaac Gym offers a more comprehensive solution by including rigid-body physics modeling, making it particularly well-suited for robotic manipulation tasks, where interaction with physical objects is essential [7].

These findings underscore the need to balance physical realism and scalability in choosing the appropriate simulation platform based on the task requirements.

TABLE I. TECHNICAL COMPARISON OF SIMULATION PLATFORMS IN EMBODIED AI

Simulator	Physics accuracy	Rendering Quality	Parallelism
MuJoCo	High	Medium	Low
PyBullet	Medium	Medium	Medium
Isaac Gym	Medium-High	Medium	High
Habitat 2.0	Medium	High	Medium

Table 1 shows the comparison of major simulation platforms used in embodied AI research across four key dimensions: physics accuracy, rendering quality and parallelism. The table highlights the trade-offs among simulators such as MuJoCo, PyBullet, Isaac Gym, and Habitat 2.0.

Simulation platforms differ not only in their technical characteristics but also in their practical applicability across

task domains. For navigation tasks, platforms like Habitat—particularly when paired with the HM3D dataset [10]—offer visually diverse and photorealistic indoor environments, making them ideal for visual navigation and path planning research. In contrast, for manipulation tasks, Isaac Gym stands out due to its GPU-based parallelism [2], which enables efficient reinforcement learning for robotic arms and supports large-scale policy training with faster convergence.

From a cost-efficiency perspective, PyBullet is often favored by startups and academic labs for rapid prototyping, thanks to its lower computational requirements and ease of deployment. Meanwhile, industries such as autonomous driving may prefer MuJoCo [9], which provides high-fidelity simulations suitable for applications that demand precise physical modeling, such as crash testing. However, a persistent challenge in this area is the lack of standardized evaluation benchmarks. As noted in comparative studies [8], cross-platform benchmarking is essential for unifying performance metrics and improving the comparability and generalizability of simulation-based research.

TABLE II. TASK-ORIENTED COMPARISON OF SIMULATION PLATFORMS AND DATASETS IN EMBODIED AI

Simulator + Dataset	Task Type	Advantages	Suitable Scenarios
Habitat + HM3D	Navigation	High-quality visual rendering; diverse environments	Embodied navigation, vision-and-language tasks
Isaac Gym	Manipulation	GPU-accelerated parallel simulation; efficient RL training	Large-scale robot manipulation and control
PyBullet	Manipulation	Lightweight, easy to use, and deploy	Prototyping for startups, academic research
MuJoCo	Manipulation	High physical accuracy; stable dynamics	Precision simulation (e.g., collision testing in industry)

Table 2 presents a task-specific comparison of simulation platforms and associated datasets commonly used in embodied AI. It summarizes their advantages and suitable application scenarios across navigation and manipulation domains.

B. Sim2Real Transfer Learning Methods

Domain Randomization (DR) [5] (e.g., randomizing friction coefficients in PyBullet [15]) is effective in static settings but struggles with dynamic conditions, such as variable lighting in warehouses. Domain Adaptation (DA) [12] uses real-world data to fine-tune models trained in simulation. These are promising for dynamic environments but require extensive and costly real-world data collection.

Techniques such as meta-learning [6] and Sim-to-Sim transfer [13] combine the strengths of DR and DA. For example, training across randomized physics parameters enabled Tobin et al. [5] to achieve 80% real-world grasp success. Table 3 presents a task-specific comparison of simulation platforms and associated datasets commonly used in embodied AI.

The choice of the Sim2Real transfer method often depends on the availability of real-world data and the complexity of the deployment environment. In data-scarce domains, such as space robotics, where collecting physical data is costly or impractical, Domain Randomization (DR) [5] offers a clear advantage by enabling robust policy training purely in simulation. In contrast, Domain Adaptation (DA) [12] is better suited for data-rich fields like autonomous driving, where abundant real-world datasets are available to fine-tune models trained in simulation.

In dynamic or high-variability environments, such as drone navigation in changing weather conditions or human-robot interaction scenarios, hybrid methods, which combine reinforcement learning with DA or DR, are particularly effective, as they balance robustness with adaptability.

TABLE III. COMPARISON OF SIM2REAL TRANSFER METHODS IN EMBODIED AI

Method Type	Advantages	Limitations	Scenario
Domain Randomization (DR)	Requires no real-world data; good for static environments	Poor adaptability to dynamic or unforeseen real-world changes	Data-scarce domains (e.g., space robotics)
Domain Adaptation (DA)	Strong adaptability to dynamic and complex environments	Relies on expensive real-world data collection	Data-rich domains (e.g., autonomous driving)
Hybrid Methods	Combines strengths of DR and DA; improves generalization	Implementation complexity; may still need some real data.	Dynamic environments (e.g., drones)

Table 3 summarizes the advantages, limitations and suitability of various Sim2Real transfer methods—Domain Randomization (DR), Domain Adaptation (DA), and hybrid methods—based on real-world application scenarios. It highlights insights into method selection depending on data availability and environmental variability.

C. Embodied Large Models

LLMs as Task Planners. Vision-language models such as PaLM-E [3] can interpret high-level instructions (e.g., “peel the banana”), but they are less effective at executing fine-grained motor control. Similarly, RT-2 [4] demonstrates the ability to understand natural language commands and convert them into corresponding robotic actions, such as “move next to the Coke can.”

Multimodal Generalization. Large language models (LLMs) support Sim2Real transfer by translating abstract instructions (e.g., “Tidy the room”) into structured sub-tasks grounded in simulation. However, semantic ambiguity remains a challenge—for example, vague commands like “clean” may lead to unintended robotic behaviors [16].

Large Language Models (LLMs) are fundamentally reshaping robotic system design. Traditional approaches follow a modular pipeline architecture — perception → planning → control—where each component is developed and tuned separately, requiring significant engineering effort and

system integration. In contrast, LLMs enable end-to-end learning, directly mapping natural language instructions to robot actions and sensory understanding, thereby unifying task-level reasoning into a single model. This shift greatly improves generalization and reduces development complexity. However, simulated environments remain essential for grounding physical intuition during training. In latency-sensitive applications like drones or mobile robots, the deployment of LLMs is still constrained by inference delay and hardware limitations. To address this, researchers have explored model distillation and compression (e.g., TinyLLaMA, DistilBERT) to facilitate efficient on-device inference.

Compared to traditional methods that require manual programming of perception, planning, and control modules, vision-language models like PaLM-E significantly simplify robot programming. Users can issue high-level natural language commands (e.g., “put the cup on the table”) without writing low-level code, reducing development time and debugging effort [3]. However, this efficiency comes with potential risks. Plans generated by LLMs may inherit societal biases embedded in the training data (e.g., gender stereotypes). Therefore, in high-stakes applications such as healthcare and education, it is essential to incorporate synthetic data audits and bias detection frameworks to ensure fairness and ethical compliance.

IV. SYNERGIES AND CROSS-DISCIPLINARY ADVANCEMENTS IN SIM2REAL TRANSFER

A. Synergies

The integration of simulation platforms and large language models (LLMs) creates a powerful synergy that enhances both perceptual grounding and action generation. For example, Habitat’s [1] photorealistic visual environments strengthen the visual understanding of models like PaLM-E [3], allowing robots to interpret and respond to complex scenes with greater accuracy. In parallel, Isaac Gym [2] provides high-performance physics simulation that enables the learning of precise motion control, resulting in a comprehensive training framework that combines semantic reasoning with physical embodiment. Another effective synergy lies in the combination of Domain Randomization (DR) and Domain Adaptation (DA). DR [5] increases robustness by exposing models to a wide range of simulated variations, while DA [12] improves task-specific precision by fine-tuning with real-world data. Together, these methods are especially powerful in long-tail scenarios such as rare object manipulation, where DR ensures environmental diversity during training and DA sharpens real-world applicability.

B. Cross-Disciplinary Advancements

Cross-disciplinary integration is accelerating the development of scalable and intelligent robotic systems. In robotics, platforms like NVIDIA Omniverse exemplify the fusion of physics simulation, visual perception, and language understanding into unified Sim2Real pipelines. This comprehensive approach enhances robotic versatility across a variety of deployment environments, from industrial automation to household assistance. In the realm of AI ethics,

as Sim2Real methods become more prevalent, it is essential to incorporate bias mitigation strategies directly into synthetic training workflows. For example, fairness assessments on datasets like HM3D [10] are necessary to ensure that the resulting models do not inherit or amplify social biases. This is particularly critical as AI-driven robots become increasingly involved in socially sensitive contexts.

C. Real-World Applications

Sim2Real technologies are finding impactful applications across multiple industries. In healthcare, surgical robots such as the da Vinci system trained via simulation provide a clear example of real-world benefits—they reduce reliance on cadavers for training, offering a safer and more ethical way for medical professionals to practice complex procedures. In terms of sustainability, GPU-accelerated simulators like Isaac Gym [2] significantly lower the computational burden of reinforcement learning training, thereby reducing energy consumption and associated carbon emissions. This shift toward environmentally responsible AI development aligns with global sustainability goals, demonstrating that high-performance learning and ecological considerations can coexist in modern robotics research.

V. CONCLUSIONS

This review has examined key aspects of embodied AI in robotics, specifically focusing on simulation platforms, Sim2Real transfer methods, and the role of large, embodied models in enhancing robotic capabilities. The key findings highlight the strengths and challenges of these areas and their interplay in shaping the future of robotics.

High-fidelity simulators, such as MuJoCo, provide unparalleled accuracy for modeling complex physical interactions but fall short in scalability, making them less suitable for large-scale training. On the other hand, GPU-accelerated platforms like Isaac Gym enable efficient, large-scale training by leveraging parallel processing, although they sacrifice some level of realism. The trade-offs between these platforms underscore the importance of selecting the right simulator based on task-specific requirements.

In terms of Sim2Real transfer, domain randomization (DR) has proven effective in static environments, enhancing robustness, while domain adaptation (DA) and hybrid methods like meta-learning show significant promise for dynamic scenarios. However, these methods often rely heavily on real-world data, posing challenges in terms of data availability and cost. Additionally, while large language models (LLMs) such as PaLM-E demonstrate impressive task planning capabilities, they face limitations in fine-grained control and real-time deployment due to their computational demands.

Despite these challenges, the future of embodied AI lies in bridging the gap between simulation and reality. There is a clear need for the development of physics-aware multimodal simulators that integrate semantic reasoning, as well as the

pursuit of lightweight, specialized embodied models capable of real-time deployment. Furthermore, the integration of ethical considerations in the design and deployment of AI systems is paramount to ensure fairness and reduce the risk of bias.

In conclusion, advancing embodied robotics will require an interdisciplinary approach, combining insights from robotics, AI, ethics, and human-centered design. By focusing on efficiency, generalization, and ethical responsibility, future research will enable the creation of robots that are not only more capable but also more adaptable and fairer in their real-world applications. The collaboration across these domains will be essential in overcoming the remaining barriers and achieving the full potential of embodied AI in robotics.

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